



Year 10 -Pathway 2
Mini Test 5.3 -Semester 2
Bivariate Data

Baldivis
Secondary College

Resources allowed: Calculator and 1 x A4 page one side

Name: Answers Class: _____ Time: 20 minutes Total Marks: ____/15

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Question 1

(2 Marks)

The following survey question is biased. Write it again to improve it:

- a) Littering is disgusting. Do you agree or do you also litter?

How do you feel about littering

✓ Or any other no biased question

The following survey question and answer set is biased. Write the question and answer set again to improve it:

- b) Pineapple belongs on a pizza (circle only one)

Never

Yuk

Gross

Or any other correct scale

Do you like pineapple on a pizza 1-5 (1=never 3=sometimes 5=always)

✓

Question 2

(Bonus point pizza on pineapple = abomination!)

(4 Marks)

For the following bivariate data please indicate which is independent (I) and dependent (D)

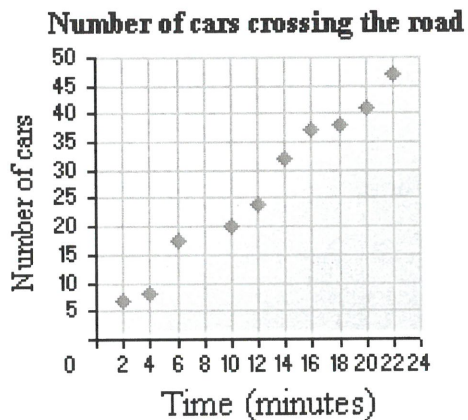
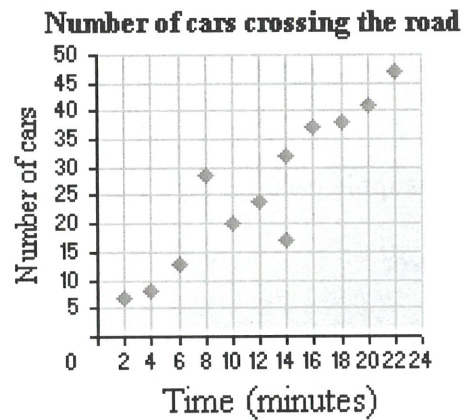
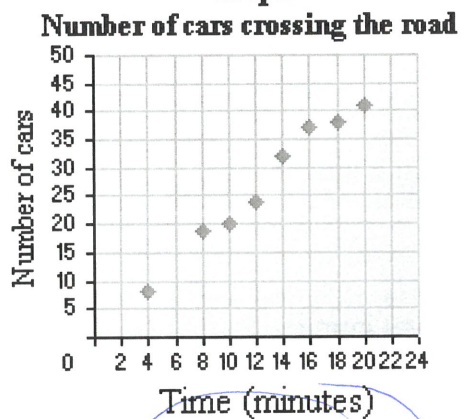
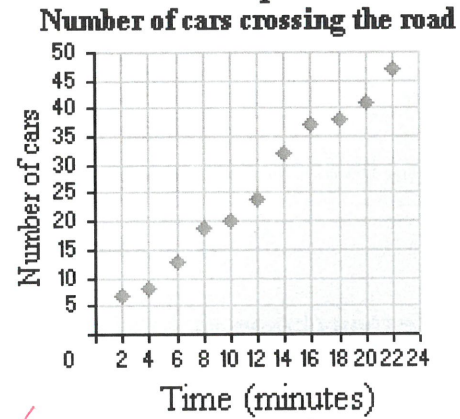
- | | | | | | |
|------------------------|----------|----|-----------------------------|----------|----------|
| a) Age | <u>I</u> | vs | Height | <u>D</u> | <i>/</i> |
| b) Distance to my home | <u>I</u> | vs | Time taken to get home | <u>D</u> | <i>/</i> |
| c) Month of the year | <u>I</u> | vs | Rainfall | <u>D</u> | <i>/</i> |
| d) Depth of bath water | <u>D</u> | vs | Time spent filling the bath | <u>I</u> | <i>/</i> |

Question 3**(3 Marks)**

Lyn counted the number of rental cars crossing on the road at two-minute intervals. The table shows the data.

Time (in minutes)	Number of cars
2	7
4	8
6	13
8	19
10	20
12	24
14	32
16	37
18	38
20	41

a) Which of the following scatter plots is correct?

**Graph 1****Graph 2****Graph 3****Graph 4**

b) Describe the correlation using terms such as positive, negative, strong, weak, non.

Strong positive

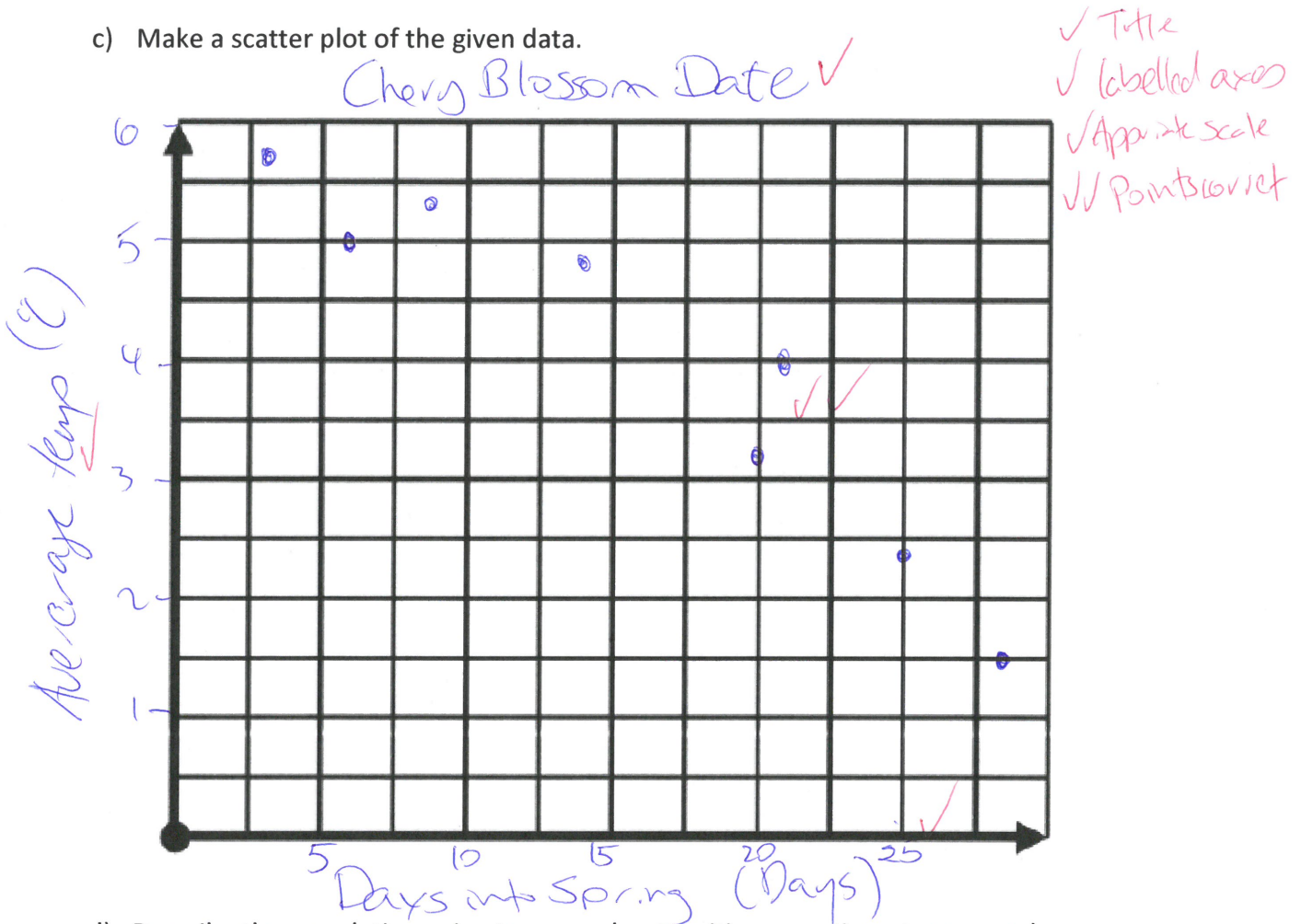
Question 4

(5, 2 – 7 Marks)

Below is the table of data regarding the cherry blossom trees in Tokyo.

Average Temp (deg C)	1.5	5.8	2.4	4.0	4.7	5.4	3.2	5.0
Days into Spring	28	3	25	21	14	8	20	6

c) Make a scatter plot of the given data.



d) Describe the correlation using terms such as positive, negative, strong, weak, non.

Negative, Strong

~ End of Test ~

